Bedside ultrasound in the diagnosis of nonalcoholic fatty liver disease

Non-alcoholic fatty liver disease (NAFLD) is the most common liver disease in the United States. While the American Association for the Study of Liver Diseases guidelines define NAFLD as hepatic steatosis detected either on histology or imaging without a secondary cause of abnormal hepatic fat accumulation, no imaging modality is recommended as standard of care for screening or diagnosis. Bedside ultrasound has been evaluated as a non-invasive method of diagnosing NAFLD with the presence of characteristic sonographic findings. Prior studies suggest characteristic sonographic findings for NAFLD include bright hepatic echoes, increased hepatorenal echogenicity, vascular blurring of portal or hepatic vein and subcutaneous tissue thickness. These sonographic characteristics have not been shown to aid bedside clinicians easily identify potential cases of NAFLD. While sonographic findings such as attenuation of image, diffuse echogenicity, uniform heterogeneous liver, thick subcutaneous depth, and enlarged liver filling of the entire field could be identified by clinicians from bedside ultrasound. The accessibility, ease of use, and low-side effect profile of ultrasound make bedside ultrasound an appealing imaging modality in the detection of hepatic steatosis. When used with appropriate clinical risk factors and steatosis involves greater than 33% of the liver, ultrasound can reliably diagnose NAFLD. Despite the ability of ultrasound in detecting moderate hepatic steatosis, it cannot replace liver biopsy in staging the degree of fibrosis. The purpose of this review is to examine the diagnostic accuracy, utility, and limitations of ultrasound in the diagnosis of NAFLD and its potential use by clinicians in routine practices.